

Abdelrahman Taha

Cairo, Egypt | tahaabdelrahmanahmed@gmail.com | +201090096410

abdelrahmantaha.me |  ORCID | github.com/taha-at | linkedin.com/in/abdelrahmanahmednageeb

Research Interests

Structural-function Mapping, Brain Connectomics, Neuroimaging (dMRI/fMRI), Neurosurgery Stereotactic Systems, AI/ML, Computational Neuroscience, and Biomedical Machine Learning.

Education

Zewail City of Science and Technology, BS in Communications and Information Engineering Oct 2023 – Jul 2027

- GPA: 3.8078/4.0 (Unofficial Transcript)
- **Relevant Coursework:** Deep Learning & Neural Networks, Linear & Non-Linear programming, Machine Learning, Probability & Stochastic Processes, Statistical Inference & Data Analysis, Digital Signal Processing, Information Theory & Coding, Communications Theory, Operating Systems

City Language School, High School – Mathematics Division (Thanaweya Amma) Oct 2020 – Aug 2023

- Grade: 96.46% – ranked 360 out of 98,685 students nationally (Transcript)

Publications & Preprints

A. Taha, Y. Mohsen, Y. Mansour, A. N. Taha, A. Abdulwahab. “Development of Abdelwahab-Taha Stereotactic Software for Neurosurgery Surgical Planning”. *Research Square* (2026). Under review at *Scientific Reports*. [Preprint]

Research Experience

Undergraduate Research Assistant, Zewail City of Science and Technology Sep 2025 – Present
Department of Mathematics **Advisor:** Dr. Abdallah Aboutahoun

- Built Python pipelines using NiBabel and SciPy to reconstruct structural connectomes from dMRI (NIFTI/CIFTI) and extract functional connectivity from resting-state fMRI across 50+ subject datasets.
- Implemented graph-based PDE diffusion models on connectome Laplacians to predict functional connectivity from structural topology, reproducing and extending published benchmarks.
- Applied Network Control Theory to quantify regional controllability and minimum-energy control trajectories between brain network states, enabling comparison across health and disease cohorts.
- Investigated relationships between diffusion dynamics, network topology, and controllability measures to identify potential biomarkers of neurological conditions.

Biomedical Sciences Lab **Advisor:** Dr. Eman Badr

- Preprocessed and analysed a corpus of 10,000+ antimicrobial peptide sequences using Pandas and NumPy, enabling systematic hypothesis testing on peptide efficacy.
- Applied XGBoost and Random Forest classifiers with Variance Threshold, Boruta, and Recursive Feature Elimination strategies, improving model interpretability and predictive performance.
- Developed modular Python pipelines for data cleaning and feature engineering, standardising the workflow and reducing preprocessing time for new datasets.

Projects

ATStereo: Stereotactic Neurosurgery Planning Software Zenodo / GitHub
Python, Stereotactic Neurosurgery, CT/MRI Registration

- Designed a source available stereotactic planning system supporting frame localization, coordinate transformation, and multi-planar trajectory calculation for Deep Brain Stimulation (DBS).

- Evaluated workflow duration across phantom CT datasets with 5 operators (3 novices, 2 experts); validated clinical reliability on Parkinson's disease DBS imaging cohorts from OpenNeuro (ds005906).
- Released open-source package on Zenodo supporting a manuscript under review at *Scientific Reports*.

MLE Estimation & Estimator Distribution in Linear Regression

GitHub

Python, NumPy, SciPy, Matplotlib, Scikit-learn

- Derived closed-form MLE estimators for OLS and WLS Gaussian linear models from first principles, including proofs of unbiasedness and covariance derivations via the CRLB and Gauss–Markov theorem.
- Ran Monte Carlo simulations (2,000 repetitions, $N=500$) to empirically validate sampling distributions; confirmed theoretical covariance $\sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$ to within 1% and demonstrated WLS efficiency gains of $2.6\text{--}3.8\times$ over OLS under heteroscedasticity.
- Applied OLS and WLS to the Engel dataset (235 households); used Breusch–Pagan testing ($LM=109.26$, $p\approx 0$) to detect heteroscedasticity and quantified the CI vs. PI trade-off across sample sizes.

2D Navier–Stokes Solver: Lid-Driven Cavity Flow

GitHub

MATLAB, Numerical Methods

- Implemented a fractional step method solver using finite-difference discretisation with iterative pressure–velocity coupling to ensure numerical stability.
- Validated convergence against published benchmarks by analysing velocity fields, streamlines, and pressure evolution under varying Reynolds numbers and boundary conditions.

DCT-Based Video Compression System

GitHub

Python, NumPy, Matplotlib

- Designed a DCT-based compression pipeline with inter-frame motion estimation (block-matching SAD), parametric quantisation (AVC and HVS matrices), and entropy coding (Huffman and arithmetic).
- Ran 5 controlled experiments varying block size, coding method, quantisation matrix, and search range; identified optimal configuration achieving PSNR of 29.33 dB at a $120\times$ compression ratio.

Audio Signal Denoising via DSP

GitHub

MATLAB, Signal Processing Toolbox

- Designed and evaluated FIR filters (Kaiser, Hanning, Hamming, Rectangular windows) to remove a 15.2 kHz sinusoidal interference; selected Kaiser Window for optimal stop-band attenuation under latency constraints.
- Estimated power spectral density using Welch's method and verified interference suppression by comparing pre- and post-filter spectra.

Courses & Training

Computational Neuroscience

Jul 2026

Neuromatch Academy (NMA) – Remote

Certificate

- Completed a 128-hour intensive interactive course focusing on computational modeling, dynamical systems, stochastic processes, and machine learning.
- Analysed task fMRI data from the Human Connectome Project (HCP) to investigate the role of the Hurst exponent as a mediator for Excitation/Inhibition (E/I) balance.
- Demonstrated that a higher Hurst exponent correlates with increased neural inhibition and stronger network connectivity.

Brain Summer School: EEG & Brain-Computer Interfaces

Aug – Sep 2026

AASTMT – Alexandria

Certificate

- Completed an intensive 1-week hands-on training on neural signal acquisition, BCI experimental paradigms, and closed-loop EEG decoding.
- Gained practical experience with live 4-channel EEG streaming (Muse 2 over LSL), signal preprocessing (band-pass filtering, ICA artifact removal), and Welch PSD feature extraction.
- Evaluated machine learning pipelines (Shrinkage-LDA, Linear SVM) for cognitive state classification, analyzing performance trade-offs between offline models and real-time streaming constraints.

Courses: Machine Learning Specialization – DeepLearning.AI (Coursera), Python Data Structures (Coursera), Learning How to Learn (Coursera)

Technical Skills

Neuroimaging & Connectomics: NiBabel, dMRI/fMRI (NIFTI/CIFTI), 3D Slicer, MRtrix, structural & functional connectomes

Modelling & Scientific Computing: Network Control Theory, PDE diffusion models, dynamical systems, SciPy, MATLAB, numerical methods

Signal Processing & BCI: EEG (Muse 2, LSL), ICA, Welch PSD, FFT, FIR/IIR filter design, DSP theory

Machine Learning & Data: Python (NumPy, Pandas, Scikit-learn), XGBoost, Random Forest, Boruta, feature selection, Matplotlib

Software & Open Science: C++ , Git, Zenodo, OpenNeuro, SQL, ASP.NET, Django, Flask

Work & Teaching Experience

Junior Teaching Assistant – Database Management Systems, Zewail City of Science and Technology 2024 – 2025

- Facilitated lab sessions for 30+ students on SQL querying and ASP.NET development; provided one-on-one debugging support that improved average assignment completion rates.

AI Trainer (ML Reasoning), Outlier – Remote Apr 2025 – Present

- Evaluated LLM outputs on advanced mathematics and programming tasks (C++ , Python), assessing logical coherence, algorithmic correctness, and step-by-step reasoning quality.
- Contributed structured feedback to RLHF training cycles, improving model accuracy on competitive-programming-style prompts.

Networking Intern, Banque Misr – Cairo Jul 2025

- Analysed enterprise network architecture applying OSI model principles and BGP/IBGP routing protocols to support infrastructure planning.

Volunteering

Organiser, RiseUp Summit – Cairo Apr – May 2025

- Organised B2B networking sessions, mentorship clinics, and investor matching events for 200+ startup founders and investors.

Organiser, TEDx ZewailCity UST – Giza Nov 2023 – May 2025

- Coordinated speaker outreach, event logistics, and technical setup across multiple TEDx events, managing a cross-functional volunteer team of 20+.